

PALU MUTIARA, Indonesia

WMO: 970720

Lat: 0.9159S

Lon: 119.9053E

Elev: 6

StdP: 101.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.3	22.8	18.4	13.3	29.6	19.4	14.1	29.1	9.1	30.9	8.3	31.4	0.4	130	0.338

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	8.6	34.8	25.2	34.2	25.2	33.7	25.1	26.6	32.0	26.3	31.5	26.1	31.3	5.9	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.2	20.4	28.3	25.0	20.1	28.1	24.7	19.8	28.0	83.1	32.3	81.9	31.9	80.8	31.4	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.9	7.0	DB	20.9	36.3	0.9	0.9	20.3	37.0	19.8	37.5	19.3	38.0	18.7	38.6	(4)
(5)			WB	19.0	27.9	1.2	1.0	18.1	28.6	17.4	29.2	16.8	29.8	15.9	30.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.0	27.7	27.8	28.2	28.3	28.4	27.8	27.6	27.9	28.2	28.4	28.2	28.0	(6)	
	DBStd	1.13	1.08	1.15	1.19	1.11	1.02	1.07	1.12	1.13	1.03	1.09	1.06	1.17	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6586	549	499	563	549	569	535	547	555	546	571	546	557	(10)	
	CDD18.3	3544	291	266	304	299	311	285	289	297	296	312	296	299	(11)	
	CDH23.3	36094	2866	2642	3126	3138	3271	2794	2759	2939	3063	3343	3082	3072	(12)	
	CDH26.7	13897	1011	978	1224	1237	1296	1029	1013	1146	1250	1373	1190	1149	(13)	
(14) Wind	WSAvg	2.1	2.1	2.2	2.3	2.1	1.9	1.8	1.9	2.1	2.2	2.2	2.2	2.0	(14)	
(15) Precipitation	PrecAvg	1215	107	83	96	114	124	120	127	93	72	91	96	103	(15)	
	PrecMax	1699	245	163	219	280	212	253	282	298	137	250	177	288	(16)	
	PrecMin	829	59	43	45	51	63	29	40	12	5	25	50	34	(17)	
	PrecStd	211	47	31	42	44	39	43	60	68	35	52	30	60	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	34.7	35.1	34.9	34.9	34.4	34.2	34.7	35.0	35.2	34.6	34.7	(19)	
		MCWB	25.5	24.9	25.0	25.4	25.7	25.2	25.2	25.0	25.0	25.0	25.2	25.4	(20)	
	2%	DB	33.0	33.5	34.0	33.8	33.7	33.3	33.1	33.7	34.0	34.2	33.7	33.5	(21)	
		MCWB	25.0	25.0	25.0	25.3	25.7	25.2	25.1	24.8	24.9	25.1	25.2	25.1	(22)	
	5%	DB	32.1	32.5	33.0	32.9	32.9	32.4	32.2	32.8	33.1	33.4	32.9	32.5	(23)	
		MCWB	24.9	24.9	24.9	25.2	25.6	25.2	25.0	24.8	24.8	25.1	25.2	25.0	(24)	
	10%	DB	31.2	31.4	32.0	31.9	32.0	31.4	31.2	31.7	32.1	32.3	31.9	31.5	(25)	
		MCWB	24.9	24.8	24.9	25.1	25.4	25.1	24.9	24.7	24.8	25.0	25.1	25.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.4	26.5	26.5	27.0	26.7	26.6	26.4	26.7	26.6	26.5	26.5	(27)	
		MCDB	31.7	31.8	32.3	31.8	32.4	31.7	31.8	31.9	32.2	32.4	32.2	31.9	(28)	
	2%	WB	25.9	25.8	25.9	26.1	26.5	26.2	26.0	25.9	26.0	26.1	26.0	26.0	(29)	
		MCDB	31.0	31.0	31.4	31.2	31.6	30.8	31.1	31.2	31.5	31.7	31.4	31.3	(30)	
	5%	WB	25.5	25.5	25.6	25.8	26.1	25.8	25.6	25.5	25.6	25.7	25.7	25.6	(31)	
		MCDB	30.3	30.4	30.7	30.6	30.9	30.2	30.3	30.5	30.8	31.0	30.7	30.6	(32)	
	10%	WB	25.2	25.1	25.2	25.5	25.8	25.5	25.2	25.2	25.2	25.4	25.4	25.2	(33)	
		MCDB	29.6	29.8	30.1	30.1	30.3	29.7	29.6	29.9	30.2	30.5	30.1	29.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	8.2	8.1	8.2	7.8	8.0	8.5	8.7	8.6	8.1	7.7	(35)	
		MCDBR	8.9	9.3	9.6	9.2	9.3	9.4	9.3	9.8	9.9	9.9	9.2	9.0	(36)	
		MCWBR	2.9	3.0	3.0	2.7	3.0	3.0	3.0	3.1	3.2	3.1	2.7	2.8	(37)	
	5% WB	MCDBR	7.9	8.3	8.5	8.2	8.3	7.9	8.2	8.5	8.8	8.7	8.3	8.1	(38)	
		MCWBR	3.0	3.1	3.0	2.8	3.0	3.0	3.1	3.1	3.3	3.1	2.8	2.9	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.418	0.425	0.425	0.418	0.418	0.416	0.424	0.459	0.491	0.554	0.474	0.428	(40)	
	taud		2.449	2.422	2.427	2.440	2.456	2.483	2.458	2.341	2.240	2.038	2.279	2.420	(41)	
	Ebn at Noon		913	910	901	886	862	851	849	838	831	787	855	900	(42)	
	Edh at Noon		118	123	122	117	111	105	109	127	145	178	139	121	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.82	5.13	5.27	5.24	5.00	4.69	4.88	5.24	5.64	5.40	5.14	4.79	(44)	
	RadStd		0.29	0.37	0.34	0.25	0.21	0.29	0.41	0.35	0.32	0.33	0.23	0.34	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.40	+0.26	+0.77	N/A	+0.42	+0.37	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.40	+0.26	+0.77	N/A	+0.42	+0.37	N/A	N/A	N/A	N/A

Nomenclature: See separate page